EX 3

Structural design III - HS 2022

Analysis and development of structures in timber

Name: _____

Introduction

Objective

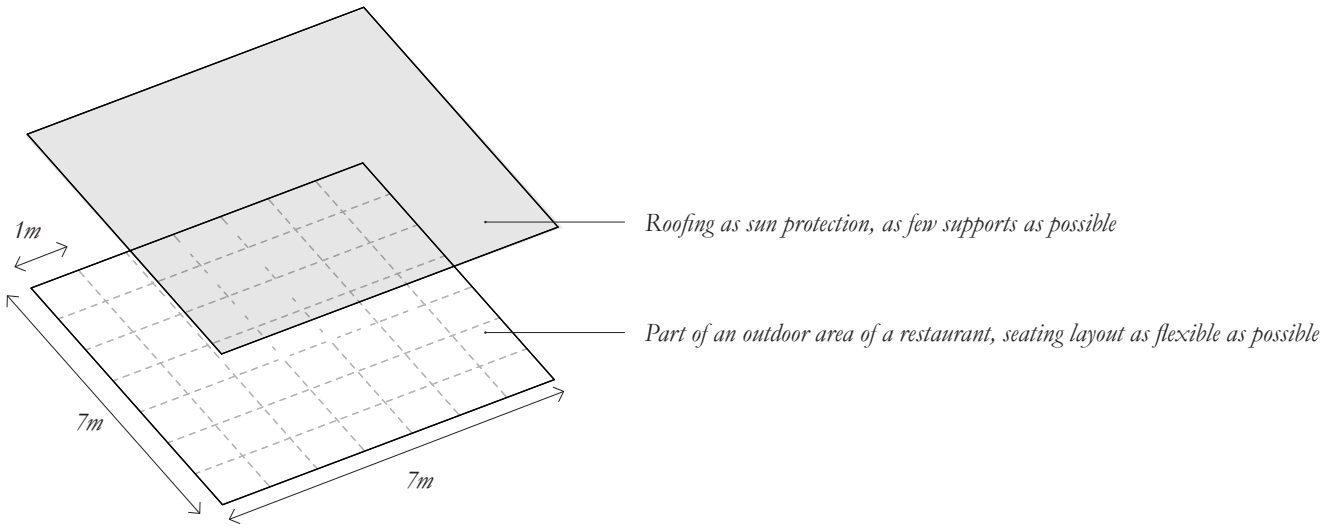
The exercises of Structural Design III deepen the understanding of the relationship between force, form and material with the help of graphic statics. These exercises illustrate the design process of a structure built from one of the main building materials: steel, reinforced concrete, timber and masonry. We will look at the following design steps: brief discussion of the concept, calculation of the relevant loads, global equilibrium, internal force flow, variations of typology-topology-geometry, design for both vertical and horizontal loads and material properties.

For further reading, following literature is recommended:

- Script «Tragwerksentwurf III-IV», Seite 43 - 54: Erläuterungen zum Konstruieren mit Holz
- Script «Tragwerksentwurf III-IV», Seite 167 - 181: Materialanhang zu Holz
- Faustformel, Seite 80 - 84: Erläuterungen zum Werkstoff Holz

Architectonical conditions

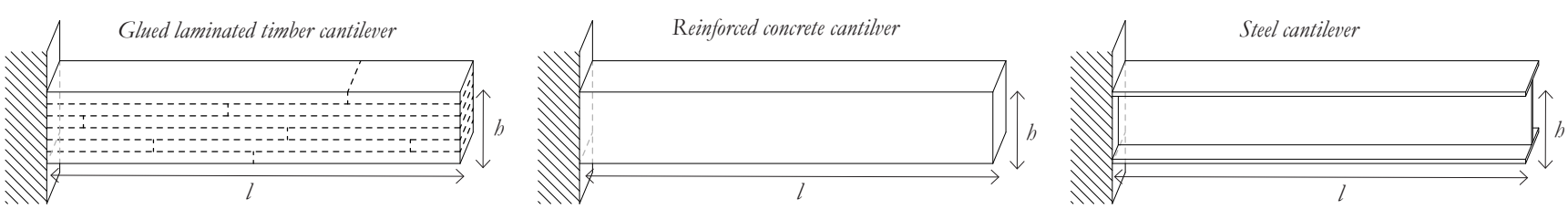
The following architectonical conditions are given: 7*7 meters of the outside area of a restaurant should be covered with a light roof to provide shading. The roof structure should be made of wood and supported by as few columns as possible to grant flexible seating.



Pre-dimensioning of cantilevers

The table on page one of the appendix «Lasten und Kennwerte» shows economical values of slendernessratios for different kinds of structures. In the previous exercise, these have already been applied to reinforced concrete beams and plates and are in this exercise adapted to cantilever beams.

In the following, cantilevers with the same dimensions, however built out of different materials, are shown. Estimate the cantileverheights h needed, in order to span a roof structure over the entire restaurant area while avoiding the use of columns. For this, the ratios for beams shown in the appendix have to be adapted in the right manner.



Hint: Anhang «Lasten und Kennwerte» S.1 →

Working with projections and three-dimensional structures

Hint: Exercises SD 3 on Steel & Reinforced concrete →

Develop an initial design for the roof. Simplified external forces are used for this gluelam structure and other external forces can be neglected. Demonstrate amount of force on tributary areas 1, 2, 3, 4 (drawing at right) ; and then use a force-diagram to determine the magnitude of the internal forces for the structural elements and given load case shown below (section at the bottom left). Use the colours red for tension, blue for compression and green for the support forces.

